Teaching Ourselves to See: A Direct Method for Denoising CRISM Hyperspectral Data

Robert Platt, Zayad AlZayer, Rossella Arcucci, and Cédric M. John

***Abstract*—The use of hyperspectral data from the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) has uncovered a wide range of previously undetected minerals on the surface of the red planet and has been instrumental in the selection of multiple rover mission landing sites. Yet, due to the aging of the sensor and an increase in noise, a significant portion of CRISM data is deemed too degraded and remains unanalyzed. We present a new data-driven model architecture, Noise2Noise for Mars (N2N4M), designed specifically to denoise hyperspectral data. Our model does not require zero-noise target data, making it well-suited for use in planetary science applications where such data does not exist. We demonstrate its strong performance over benchmark methods, outperforming them on most metrics. We suggest that the reliance on spatial correlations in noise profiles for the deep learning (DL) benchmark methods may be part of the cause of their poor performance when applied to CRISM data. We apply our methodology to two case study sites, each previously studied for hydrated mineralogy, and demonstrate that our model can increase the visual clarity of summary parameters used to highlight areas of interest. This has potential impacts for the characterization of landing sites for upcoming missions such as the ESA's ExoMars rover mission.**

***Index Terms*—Denoising, hyperspectral, machine learning (ML), Mars.**

I. Introduction

THE mineralogical composition of the surface of Mars contains a plethora of information about the history and development of the red planet [1]. Surface missions provide finely detailed information of localized areas, but satellite remote sensing data is relied on to provide the bulk of information about the planetary surface. The Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) provided the highest resolution hyperspectral imagery of Mars across the visible, near-infrared (NIR), and shortwave infrared (SWIR) wavelengths, with over 30 000 images of the planet [2].

However, CRISM was decommissioned in 2022 due to the degradation of image quality over time, attributed to cryocooler failure and the increase in sensor temperature [2]. Noise in the images limited the range of possible analysis as it obscures mineral absorption features, with a noticeable impact since 2012 [3]. This potentially hinders understanding of large areas of the Martian surface that were only imaged since sensor degradation became significant. Notably, this includes Oxia Planum, the future landing site for the Rosalind Franklin rover mission of the European Space Agency (ESA). Of the 18 available CRISM images of the area, 28% (five images) are unusable due to high noise fractions [4]. The ability to denoise spectra to allow for analysis of these extra images could significantly improve our understanding of Oxia Planum before the landing of the ExoMars mission.

A number of artifacts and noise sources impact the quality of CRISM targeted observations: residual atmospheric distortions, interpolation bias, water–ice aerosol spectral effects, and random noise [5]. It is the random noise fraction that has increased significantly over time, thus preventing detailed analysis of some imagery [6]. This random additive noise is believed to be caused by increased sensor temperature, due to successive failures of the cryocoolers on board [2]. The most significant atmospheric distortions are caused by a triad of CO_2 absorption features between 1.9 and 2.1 μm [7]. These are significant as they interfere and mask the detection of hydration and surface H_2O absorption features, which are targets of interest on Mars [8]. The typical method for dealing with these atmospheric absorptions is the so-called "Volcano-Scan" correction. However, due to variability in atmospheric conditions during acquisition, some artifacts often remain or are induced by an imperfect correction [8]. In this work, we attempt to address both the additive random noise and the most common remnant atmospheric artifact, as these have the largest impact on the usability of CRISM imagery.

The handling of noisy data is one of the most common issues facing data scientists, and there are many methods of machine learning (ML) to mitigate this problem [9]. Typically, such methods are supervised ML algorithms, which rely on the availability of noisy data (our features) and denoised data (the ground truth, or target) at the same location. However, very few repeat images of the same area of the planetary surface exist, and even the earliest CRISM images still contain some noise [10]. This limits the available dataset for supervised denoising ML techniques, which require this paired input and target data. To leverage larger quantities of data, unsupervised or semi-supervised techniques have been historically preferred for CRISM applications [11], [12].

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Recent interest in the use of ML with satellite imagery of Mars has demonstrated that ML techniques can prove highly effective at processing and analyzing CRISM data. Spectral cluster maps have been used to fuse CRISM and data from the Color and Stereo Surface Imaging System (CaSSIS) [13] for better combined spectral and spatial resolutions [11]. Another clustering approach used k-means and UMAP to identify spectrally distinct regions [12]. More recently, CRISM data were combined with thermal data and grain properties and used in a support vector machine-based model by Koeppel et al. [14] to cluster Martian terrains. Generative adversarial network (GAN)-based methods have been used both for classification [15] and as a super-resolution technique [16]. With the release of the CRISM ML Toolkit Dataset [17] alongside their spectral classifier, some works have applied supervised techniques for classification, such as the convolutional neural network (CNN) developed by Kumari et al. [18]. The CRISM data for all these works is predominantly from pre-2012, when the image quality was significantly higher. Here, we focus on developing techniques to restore all CRISM data quality to levels sufficient for further analysis.

In this study, supervised ML techniques are employed in conjunction with a synthetic noise generation method, enabling the utilization of the full range of CRISM data. This approach facilitates the development of a highly effective denoising model, thereby unlocking a substantial body of previously unusable CRISM imagery for detailed scientific analysis. Notably, this includes imagery of Oxia Planum, which is of critical importance for advancing our understanding of the ExoMars landing site in preparation for that mission's launch. The methodological contributions are as follows.

1) We introduce a novel self-supervised deep learning (DL) architecture based on the Noise2Noise framework, designed specifically for hyperspectral data. We show that it achieves state-of-the-art performance in denoising CRISM spectra and release it as a benchmark for future comparison.
2) Our model is trained directly on CRISM data, where the target data is not entirely noise-free, and we demonstrate that this produces a better model than transferring a model that was trained on different sensors and datasets.
3) By utilizing 1-D convolutions, our model has a significantly smaller parameter count than other SOTA methods, supporting recent works suggesting that larger models are not a requirement to successfully leverage remote sensing data [19].

II. Related Works

A. Existing CRISM Denoising Methods

Various methods have been developed to reduce the noise and artifacts prevalent in CRISM data. The "Volcano-Scan" correction is the current method applied in the preprocessing pipeline for CRISM IR images to correct for atmospheric effects [20], [21], [22]. The Volcano-Scan method was originally developed for Observatoir pour la Mineŕálogie, l'Eau, les Glaces et l'Activite (OMEGA) data and adapted to use for CRISM imagery [23]. The method divides all spectra by a scaled empirical transmission spectrum derived from the difference in absorption between the base and summit of Olympus Mons. The original method used a single transmission spectrum; further development added additional spectra to select from based on atmospheric conditions at the time of image acquisition [8]. This does lead to increases in signal quality, but selecting the correct transmission spectra and scaling factor is nontrivial, and often remnant artifacts are observed [5]. This method is part of the publicly available CRISM Analysis Toolkit (CAT) software package and implemented on the Mars Système d'Information (MarsSI) web-GIS application [8], [24].

Several statistical denoising approaches build upon the products after the Volcano-Scan correction. The CRISM Iterative Recognition and Removal of Unwanted Spiking (CIRRUS) method is able to identify and remove outliers by measuring the signs and magnitudes of the differences in neighboring reflectance values, to "despike" and "destripe" the images [25]. Carter et al. [26] developed a tool, OCAT, which used a fast Fourier transform (FFT) to generate a smoothed signal. Outliers were identified using a combination of two criteria. First, the difference between smoothed and raw signals must be greater than a selected threshold. Second, there must be an inflection point in the raw data that is not present in the smoothed signal. Outlier values were replaced by linear interpolation between in-distribution datapoints. A third method called the Complement To CRISM Analysis Toolkit (CoTCAT) used a series of moving average filters to iteratively smooth and denoise the spectra [3].

Radiative transfer modeling offers an alternative to the Volcano-Scan and the postprocessing methods that complement it. The first approach to apply this methodology to CRISM data is discrete-ordinates-method radiative transfer (DISORT), which utilizes a physical forward model to correct for photometric, atmospheric, and absorption effects explicitly [27]. Further works built on the approach, applying maximum likelihood estimation (MLE) to the results given by DISORT [28], and adding hypothesis testing for the distribution of the data and noise [29]. Itoh and Parente [5] developed a method to directly estimate the atmospheric transmission from the image itself, to simultaneously apply atmospheric correction and denoising. These methods are potentially more accurate than the Volcano-Scan method, at the cost of computational complexity, with computation time of up to 2 h [5]. This limits the applicability of the method to the entire CRISM dataset of >30 000 images.

B. ML-Based Hyperspectral Denoising Methods

ML, specifically DL, has produced powerful methods for hyperspectral denoising over the last several years [30]. The first DL methods used 2-D CNNs [31] and were quickly developed to include residual connections [32]. A further addition was the use of 3-D rather than 2-D convolutions, which allows the model to exploit the full spatial–spectral correlation of the data, rather than solely the spatial component [33]. This can be done separately as 2-D convolutions followed by 1-D convolutions [34], or jointly as 3-D convolutions [33]. MemNet [35] uses a recurrent network structure, with

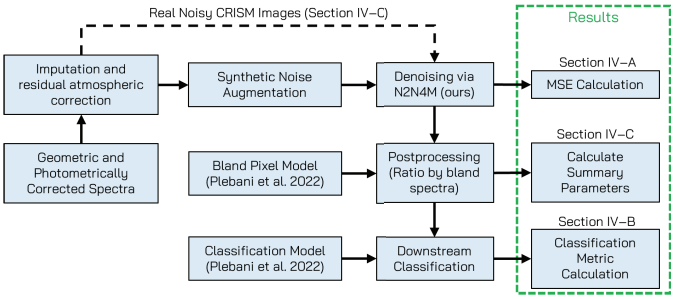


Fig. 1. Schematic flowchart showing the preprocessing, training, and evaluation steps of this work. Results sections are outlined in green. Note noise augmentation is bypassed in the deployment case, when the inputted images already contain significant noise fractions.

a memory gate to only pass useful information between units, enabling long-term dependencies. QRNN2D and QRNN3D [36] extend this to use 2-D and 3-D convolutions, respectively, in a quasirecurrent framework, with an alternating directional structure to remove causal dependency between channels.

III. Methodology

The methodology presented here consists of a sequence of preprocessing steps before denoising (Fig. 1). Evaluation is conducted on various levels of postprocessed product, to allow for thorough assessment of the denoising capability of the model, the downstream impact of denoising on spectra, and a visual assessment of result quality across whole images with real noise.

A. Model

The model we propose here, Noise2Noise for Mars (N2N4M), is a CNN in a U-Net architecture [37]. It has been shown in prior works that models implementing this architecture are capable of learning to denoise imagery, despite the presence of noise in the target imagery itself [38]. This is the basis of Noise2Noise (N2N) denoising methods. While previous research applied this methodology to denoise natural imagery [38], this is the first work to use this idea to denoise hyperspectral imagery, where the target imagery is known to be partially corrupted. This property is key for denoising CRISM spectra, where even older, high-quality data still contain some element of noise [10]. Here, we choose to apply 1-D convolutions (Fig. 2) acting over the spectral axis instead of the typical 2-D convolutions applied to the spatial axes. Spectral context contains key information to accurately recreate the target spectra, with global slope and local absorptions needed for mineral identification [39]. The large wavelength range covered by the dataset and lower spectral resolution compared to sensors such as ROSIS and CASI mean there is much less shared information between bands. Our model is available on GitHub: https://github.com/rob-platt/n2n4m, making it the first posthoc denoising method for CRISM data that is openly accessible to CRISM data users.

Previous works have leveraged high-quality laboratory spectra collected on Earth for training DL models to classify CRISM data [40]. However, there is a highly limited number and diversity of lab spectra, and critically, they often display differences in absorption features to CRISM spectra [39]. The causes for this are varied, including imperfect corrections for atmospheric and dust contributions to spectra, and potential chemical differences between "pure" lab samples and "impure," mixed spectra on Mars observed over a 324 m^2 pixel. Using lab spectra as the ground truth in our scenario could cause our model to learn undesirable properties, such as mapping the true absorption features seen in the CRISM data to match those in the lab spectra. Instead, we chose to train our model directly on CRISM data, selecting spectra from earlier, low-noise CRISM imagery. Synthetic noise is then added to these spectra to form noisy input data. This forms paired high-noise and low-noise training data for a supervised model. This circumvents the requirement for paired data or the use of lab spectra and allows for the use of a highly diverse training dataset from any CRISM imagery.

Model architectural options, such as the number of upsampling and downsampling operations, the number of convolutional layers per convolutional block, and kernel size, were all treated as hyperparameters and tuned as such. Standard hyperparameters such as batch size, number of epochs, and learning rate were also included. The hyperparameter search was accomplished using the Raytune library, with a random search over the hyperparameter space. Early stopping was implemented such that if there was no improvement in mse of the validation dataset over 10 epochs, the training process was ended. The full hyperparameter space and final model specification can be found in the Appendix. Convolutional kernel size was found to have a significant effect on results, with larger kernel sizes producing results with more high-frequency noise. This leads to the recommendation that future works also experiment with this parameter when applying DL methods to spectral data from other sensors, as the best performance is likely linked to the spectral resolution of the sensor and the SNR of the individual dataset. Batch size and learning rate were found to have the greatest effect on training stability, with best results obtained when the ratio of batch size to learning rate is low, as found in other works [41].

B. CRISM Sensor and Dataset

The CRISM instrument is composed of two separate sensors, the "S" detector observing visible/NIR (VNIR) wavelengths between 0.36 and 1.05 μm and the "L" detector observing SWIR wavelengths between 1.01 and 3.92 μm [1]. Secondary minerals on Mars, the primary targets of the sensor, have diagnostic absorption features at 1.4-, 1.9-, 2.1-, and 2.2–2.5 μm [2], thus data from the "L" detector are the focus of study for the majority of CRISM data users. As the source of noise in CRISM data is attributed to an increase in detector temperature [6], there is a more significant effect on IR wavelengths. Hence, we choose to focus our denoising efforts solely on the "L" sensor, where the noise contribution is the most significant, and clear identification of absorption features is most necessary. Data from a wide variety of hyperspectral sensors have been used to demonstrate ML applications, with a number of common sensors and derived datasets summarized in Table I. The key differences

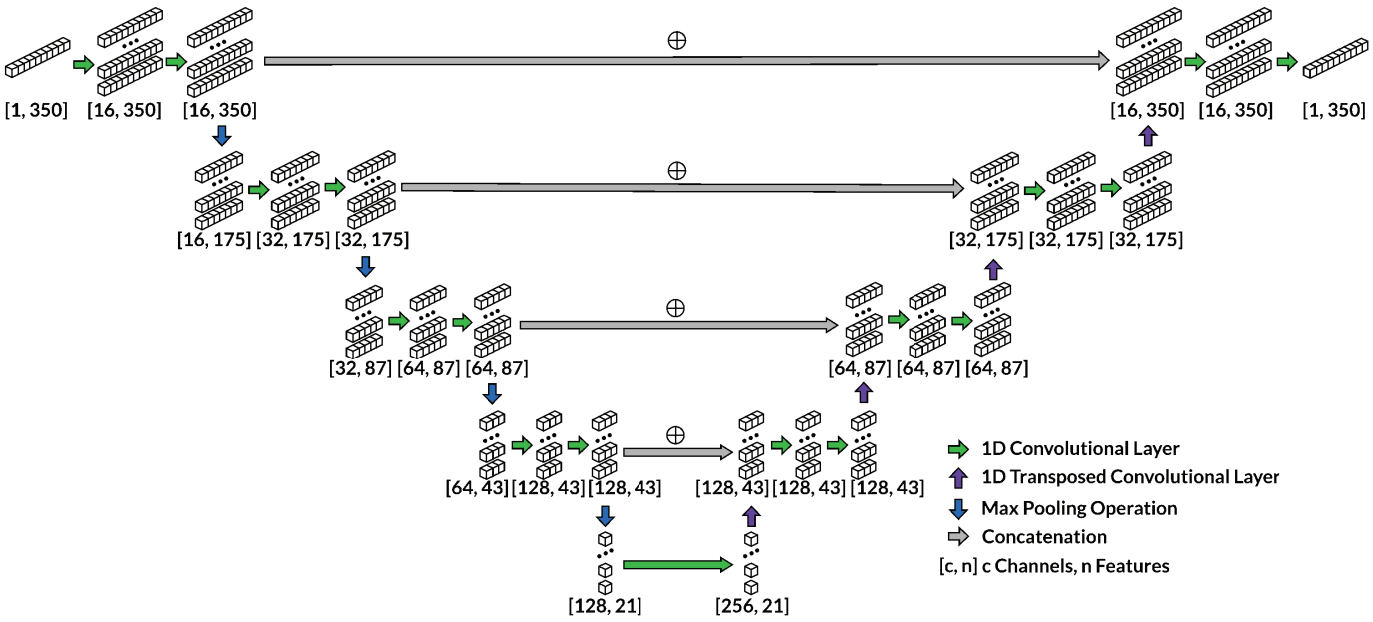


Fig. 2. N2N4M network architecture. The model uses a U-Net architecture with convolutional layers acting over the spectral dimensions of the data with a kernel size of 5.

TABLE I
COMPARISON OF COMMONLY USED HYPERSPECTRAL SENSORS WITH CRISM. A GOOD DESCRIPTION OF MANY OF THESE DATASETS CAN BE FOUND IN [42]

Sensor	Mount	Spatial Resolution	Spectral Resolution	Spectral Domain	Number of Channels	Derived Datasets
ROSIS	Airborne	1.3m	4nm	0.43um-0.86um	103	Pavia Centre/University
CASI	Airborne	2.5m	3.5nm	0.38um-1.0um	144	Houston
HYDICE	Airborne	2m	10nm	0.4um-2.5um	191	Washington DC Mall
Kappa DX4	Ground-based	Various	10nm	0.4um-1.0um	31	ICVL
AVIRIS	Airborne	20m	10nm	0.4um-2.5um	220	Indian Pines
CRISM-L	Satellite	18m	6.55nm	1.01um-3.9um	438	CRISM ML Toolkit

between sensors used previously and CRISM are that CRISM is a satellite-based sensor, requiring complex atmospheric corrections [8], [27]. In addition, CRISM L has a far larger number of spectral bands (438 bands) [1]. Both factors must be considered when designing a denoising strategy to apply to CRISM data.

For this work, data were derived from the CRISM ML Toolkit dataset [17]. This is comprised of ~600 000 pixels from 77 images acquired with the CRISM L sensor, each associated with one of 39 mineral labels. The base targeted reduced data record (TRDR) images were first processed using MarsSI [24] to apply the standard photometric and Volcano-Scan atmospheric corrections. The TRDR processing converts the radiance values to *I/F*, correcting for the solar spectral irradiance observed by MRO [2]. The Volcano-Scan correction attempts to remove the contribution of atmospheric effects from the spectral signature [8]. TRDR products are the most ubiquitous CRISM data type, as the higher-level TES and MTRDR products are only available for a small subset (~40%) of CRISM observations [2].

To ensure the strong generalization ability of the model for the intended use data, there must be minimal distribution shift between the training and testing data, and the expected use data [43]. Practically, this means that the data features, the spectral shape, absorption features, wavelength range, and noise distribution, should have similar means and variance in both the training data and deployment scenario. To guarantee this, spectra from TRDR imagery were chosen for the training data, as TRDR data are the expected use case for the model. Each IR spectra contains 350 channels between 1.0210–2.6483 and 2.8070–3.4769 μm, with channels known to contain artifacts excluded as in the original data [17]. CRISM imagery also contains a significant proportion of spectrally inert or "bland" pixels, which are not included in the initial dataset. These are important as in deployment the model will encounter such data regularly, and if not trained on them, will "hallucinate" absorption features in the noise fraction of bland spectra. We process and add to the training dataset 150 000 "bland" pixels identified by [17] from a wider range of images. This helps to regularize the model and prevent spurious absorption features from being introduced during the denoising process.

This dataset represents a more challenging learning problem than other commonly used hyperspectral datasets for ML; it has a sparse, global extent, and was acquired for 9 years, unlike datasets such as Pavia University or Washington DC Mall, which were collected in a single acquisition. Spatial autocorrelation can be a significant issue in geospatial datasets

[19], [44], but with the correct train/test split, this can be avoided by leveraging the global nature of the dataset.

C. Training, Validation, and Testing Sets

A key desired property for ML models is generalization capability to previously unseen data. This is typically assessed with the use of a test dataset, which the model does not see at training time. Many previous works across multiple domains investigate the optimal ratio of training to testing data [45] and the development of regularization methods to prevent overfitting. However, fewer works examine the validity of their choice of test dataset, which should be representative of the deployment scenario for the model. For models leveraging remote-sensing data, deployment scenarios often involve application in unseen geographic regions [46], or planets [47]. Kalantar et al. [48] showed that the geographic area used for training has a significant impact on model weights and feature importance. Despite this, it is still common practice to randomly split data to form training and testing datasets [49].

Using a separate geographic area to test model performance should ensure no spatial autocorrelation between training and testing sets [46], [50]. Defining such an area can be challenging, with [45] showing that there is a significant spatial autocorrelation even for nonadjacent areas. In [51], localized Random Forest models were trained on limited geographical areas and bagged together for predictions. While effective, this would have prohibitively large computational costs if applied to DL methods, which we would expect to be more performant on hyperspectral data [40].

Salazar et al. [52] developed a novel method for defining a spatially fair training–testing split for reservoir characterization based on geostatistical methods. Their method assumes that the training, testing, and, most importantly, the deployment scenario belong to a single, if heterogeneous, spatial area. This is not necessarily true for remote sensing-based models, which may involve deployment cases that are very spatially separate, for example, different ecological zones [44].

Although the data is inputted to the model on a spectra-by-spectra basis, the division of data into training, testing, and validation datasets was completed at an image-wide level. This ensured no geographical autocorrelation between training, testing, and validation, so that test metrics are an accurate reflection of model ability on unseen data, which in a deployment scenario would be on new images. In addition, the serpentine and muscovite mineral classes were partitioned into the test set only to provide insight into the model's ability to generalize to unseen mineral classes. This is critical for the downstream classification use case of the model, as we cannot assume that the dataset contains samples of every mineralogy we can expect to find on the surface of Mars.

A dataset of paired high-noise and low-noise data is required to train an ML model in a supervised regime. There are insufficient instances of repeat CRISM image acquisitions over a given area that have significant SNR degradation between images to curate a large training dataset from them. Instead, a large quantity of CRISM spectra were obtained and designated as the target spectra for the model. Synthetic noise was then added to each to form the paired, high-noise data. Using this framework, any CRISM L-sensor TRDR images could be used to obtain the target spectra. This curated dataset was selected for several key reasons.

1) The image quality is known to be high.
2) The mineralogy has been identified and labeled, which allows for testing of the downstream impact of denoising.
3) The images represent a broad geographic area and diverse mineralogy, which allows for a thorough assessment of the generalization capabilities of the model.

The exact images and data split used can be found in [17] and at our codebase https://github.com/rob-platt/n2n4m

D. Preprocessing

Two preprocessing steps were applied to clean the data: imputation of bad values ($I/F > 1$), and of data where there was a significant atmospheric artifact remaining between 1.91 and 2.08 μm due to issues with the Volcano-Scan correction. A linear interpolation was calculated between 1.91 and 2.08 μm and compared to the spectra over that range using the R^2 metric. If the spectra deviated sufficiently for the R^2 value to drop below 0.7, that spectrum was flagged as having an anomaly in that region. A per-band mean was calculated for spectra considered not to be anomalous, on a mineral-by-mineral basis within each image. Any anomalous spectra had the artifact replaced by this mean. By using the average given by the same mineral species within the same image, any image or mineral-specific features were retained where possible. If all spectra from a given image contained anomalies, the artifacts for these spectra were imputed by the per-band average for that mineral globally across the dataset.

To create paired high-noise and low-noise training data, synthetic noise was artificially added to the low-noise data. Computing the exact noise profile of CRISM spectra is intractable, due to the absence of noise-free ground-truth data for the surface of Mars [29]. Previous CRISM modeling and denoising works have used forms of Gaussian distributed noise to model the random noise in CRISM data with some success [53], [54]. Each hyperspectral image (HSI) cube can be denoted as a tensor $I \in \mathbb{R}^{H \times W \times C}$, with rows H, columns W, and spectral channels C. Each spectrum is then described as $x \in \mathbb{R}^C$. Synthetic noise z^C was created by sampling from a multivariate standard Gaussian distribution [3]

$$z \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{1}$$

where $\mathbf{0}$ is the zero mean vector, $\mathbf{I}$ is the identity matrix. An additional, uniformly sampled random variable λ^C is used to increase the intersample variability, which better reflects the observed true noise distribution [3]

$$\lambda \sim U(0, 1). \tag{2}$$

This gives a final noise model of

$$x_{\text{noisy}} = x_{\text{clean}} + \lambda z. \tag{3}$$

Fig. 3 shows an example of a noisy and a low-noise spectra of the same location, and a range of synthetic noise spectra. The training objective is to optimize a network g such that

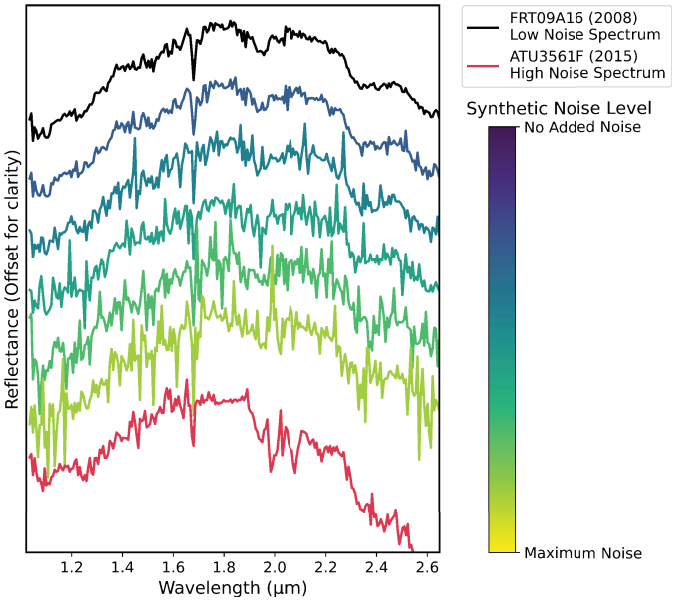


Fig. 3. Example of a low-noise spectrum from image FRT00009A16, acquired in 2007, and a high-noise spectrum from image ATU0003561F (2015) over the same area. Also shown is the low-noise spectrum with increasing levels of synthetic noise added (corresponding to lighter line colors), which comprises our training data.

$g(x_{\text{noisy}}) \approx x_{\text{clean}}$. The selected synthetic noise distribution is believed to closely simulate that of the true CRISM noise distribution [3], and so the trained model will be capable of denoising real CRISM data. This assumption is tested in Section IV-C.

E. Evaluation Methods

Model evaluation is performed in three distinct stages. In Section IV-A, quantitative denoising performance on a set of unseen test pixels with added synthetic noise is assessed. N2N4M performance is compared with a range of other state-of-the-art HSI denoising methods, using the models and pretrained weights from HyDe [55]. We use Memnet [35], and QRNN2D and QRNN3D from [36], all demonstrated to be effective methods for hyperspectral denoising. Comparisons to methods commonly used with CRISM data, Savitsky–Golay filtering [56], and CoTCAT [3] are also made.

In Section IV-B, the impact of denoising on downstream task performance is examined. The downstream use case of CRISM data after processing is automated mineral classification, so the impact of N2N4M denoising is assessed alongside the other DL and common statistical methods listed above. To conduct the automated classification, the hierarchical Bayesian model (HBM) from [17] is used, as it has the largest class range and is the most performant of current CRISM classification systems. After denoising, to prepare the data for classification, a "ratioing" step must be completed. This involves the division of spectra of interest by nearby "bland" spectra which show limited/no absorption features [1], [57]. This acts to enhance minor spectral features and allows for easier identification and classification. Bland pixels are typically selected from the same detector column as the pixel of interest, to subdue the effect of any detector-based instrument artifacts. The method used to select which pixels are bland can have a significant impact on results [3]. The most usual method is to select pixels by hand, which can introduce user bias and is time-consuming. Using a column-median as a bland or common signature [26] can be effective, but relies on the relatively small size of areas of interest compared to the overall image size; otherwise, features of interest will be suppressed [3]. In [3], they propose a "selective ratioing method" to identify bland pixels using interband variability; however, this is sensitive to broader absorption features, which may not vary significantly between neighboring bands.

Here, the ML method developed by [17] is used, which leverages an HBM to calculate the likelihood of a given pixel being bland. To further reduce any noise, the 3 pixels with the highest likelihoods per column are averaged and used as the denominator for ratioing. The resultant ratioed spectra are then used as input to the classification model. The HBM classifier was used to classify a test set of ~100 000 unseen spectra, with known mineral labels, acquired from a range of sites across the surface of Mars [17]. These labels correspond to 30 distinct mineralogies, the most diverse large-scale dataset of CRISM mineralogy collated to date [17]. These minerals represent the majority of those confidently confirmed to be present on Mars [39].

In Section IV-C, two pairs of CRISM images are introduced. In each pair, one image was acquired early in the mission and the other several years later, containing significantly more noise. N2N4M is used to denoise the high-noise image, and a qualitative comparison of the output to the low-noise image is made. To accomplish this, the low-noise, high-noise, and denoised images are individually ratioed using the method described above. To highlight regions of interest for a given mineral group in an image, summary parameters [39], [58] have been developed specifically for CRISM data. These calculate the relative reflectance across spectral regions where absorption features are expected for a given mineralogy and highlight pixels that contain those absorption features. Here, summary parameters for known mineral species are selected for each image, identified from the low-noise images in each pair. A visual assessment of the ease of identification of mineralogy from the summary parameters across the low-noise, high-noise, and denoised images is undertaken to understand the effectiveness of denoising using N2N4M.

IV. Results

A. Synthetic Results

When applied to unseen data with synthetic noise, our model demonstrates strong quantitative performance (Table II), with a lower mse than all benchmarks. The standard deviation of the mse of N2N4M is also the smallest. This is significant as when deployed, the model will be applied to denoise an entire image rather than individual pixels. Thus, a low variance in results is highly important to ensure consistency across the image. The mse on the training and validation data were 0.67×10^{-5} and 0.27×10^{-5}, respectively, indicating no significant under- or overfitting occurred during training.

TABLE II

PERFORMANCE COMPARISON WITH COMMON REMOTE SENSING AND STATE-OF-THE-ART ML DENOISING METHODS FOR BOTH SYNTHETIC DATA DENOISING AND DOWNSTREAM TASKS. DOWNSTREAM PERFORMANCE GIVEN RELATIVE TO PERFORMANCE ON LOW-NOISE DATA. THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD

Data	Denoising Task MSE ($\times 10^{-5}$)	Relative Accuracy	Relative F1 Score	Relative Precision	Relative Recall
Savitzky-Golay Filter [56]	2.8 ($\sigma = 0.6$)	0.01	0.00	0.42	0.09
CoTCAT [3]	0.5 ($\sigma = 0.15$)	0.35	0.43	**0.50**	0.41
QRNN2D [36]	0.65 ($\sigma = 0.63$)	0.08	0.1	0.38	0.18
QRNN3D [36]	0.45 ($\sigma = 0.31$)	0.16	0.19	0.42	0.18
Memnet [35]	1.3 ($\sigma = 0.35$)	0.03	0.05	0.25	0.09
N2N4M (Ours)	**0.42** ($\sigma = 0.12$)	**0.52**	**0.48**	**0.50**	**0.64**

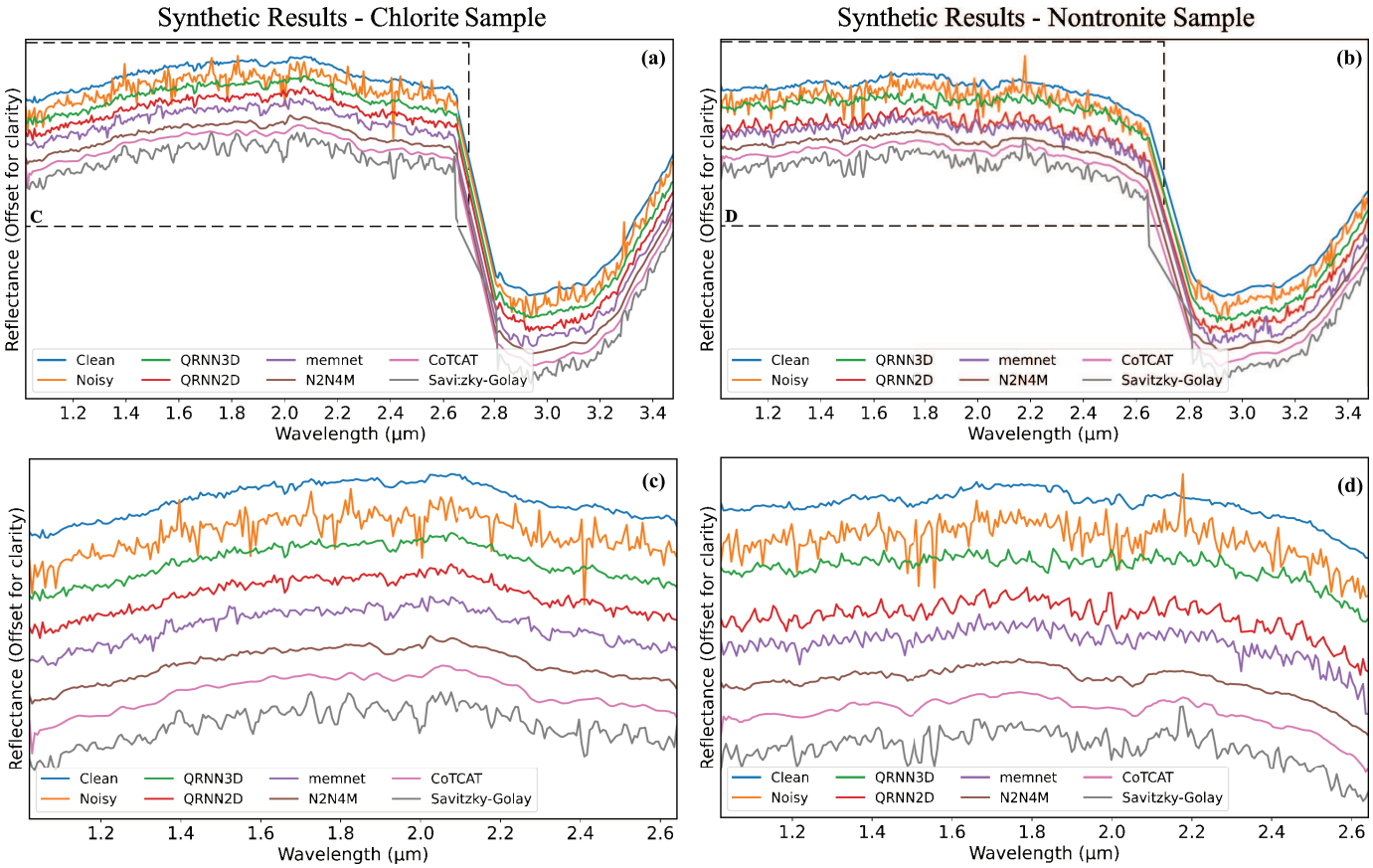


Fig. 4. Results for N2N4M and benchmark methods denoising unseen spectra with synthetic noise. (a) and (b) Sample of the minerals chlorite and nontronite, respectively. The dashed boxes outline the area plotted for (c) and (d) of the 1.0–2.6-μm range, which contains most key absorption features for mineral identification.

With a visual inspection of the results (Fig. 4), it is observed that most methods retain the overall shape of the spectra and remove the largest amplitude noise spikes. Where N2N4M outperforms other models is in the removal of lower-amplitude, higher-frequency noise from the signal.

N2N4M performance is also critically evaluated on unseen mineral classes, to ensure that the model has learned to denoise, rather than memorizing key features of target spectra. The model was tested on spectra of muscovite/illite and serpentine, hydrated minerals which are of significant scientific interest on Mars [57], [59], and detectable by automated methods using SWIR hyperspectral data [60]. As can be seen in Fig. 5, the model successfully denoises spectra to allow for visual identification of key absorption features required to uniquely identify the mineralogy. This suggests the model has successfully generalized beyond the mineralogy shown in its training data.

B. Downstream Task

The primary use case for CRISM data is to classify the mineralogy of the Martian surface, and therefore, it is essential to assess the impact of the denoising methodology on this task. In Table II, the classification metrics of these results are reported, relative to the performance of the HBM on the clean data. N2N4M shows a significant improvement in downstream task performance over benchmark approaches across all metrics. In particular, we note an increase of 0.23–0.55 (50%–610%) in relative recall, suggesting that the HBM can more consistently identify absorption features in

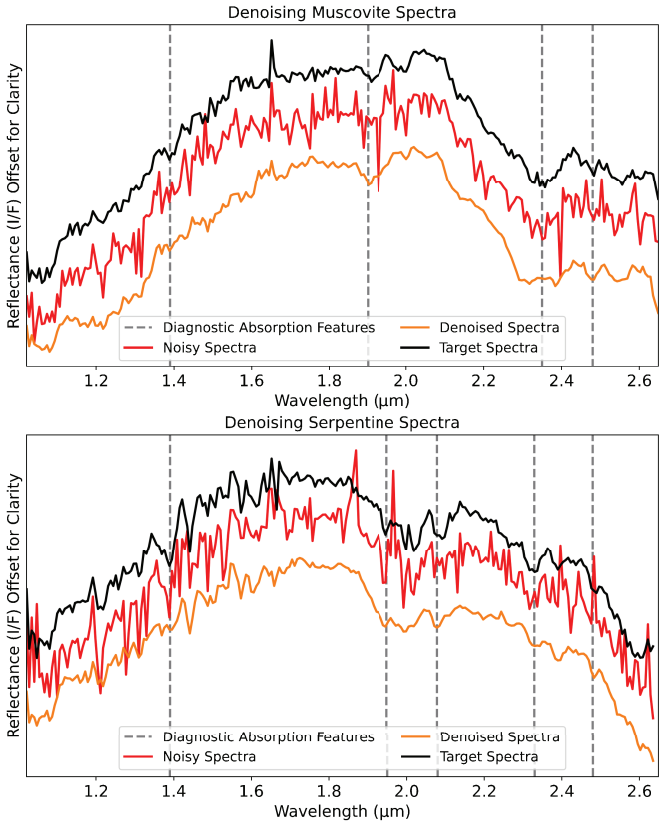


Fig. 5. Examples of low-noise, noisy, and spectra denoised by N2N4M for serpentine and muscovite, two mineral classes that the model did not see during training. N2N4M denoised outputs clearly show the key absorption features for mineral classification, despite the combination of these being specific to the mineral of interest. This shows the model has generalized well rather than memorizing specific features.

the samples denoised by N2N4M compared to other methods. This leads to more positive identifications of minerals rather than "bland" background pixels which display no features of interest, increasing model recall scores.

C. Real Image Results

For final model evaluation, we selected two test sites as case studies (Figs. 6 and 7). Each site involves a pair of CRISM images taken over the same spatial area, with multiple years between image acquisitions. Sites were selected based on the availability of repeat imagery, low dust coverage, which may vary between acquisitions, and areas of geological interest. The first pair, FRT0000B7DA and FRT000180E9, acquired in 2008 and 2010, respectively, show an area of Noctis Labyrinthus. Ca-sulfate bassanite and/or gypsum were identified in image FRT0000B7DA [61]. The second pair, FRT00009A16 acquired in 2008 and ATU0003561F acquired in 2015, shows a region of Oxia Planum, the landing site for the upcoming ExoMars mission [24]. The older FRT image has been studied in detail, with large clay deposits visible [4], [10]. Both pairs of images have previously been used to highlight the degradation of the CRISM sensor and the difficulty of making positive mineral identifications in noisier CRISM images [3], [10].

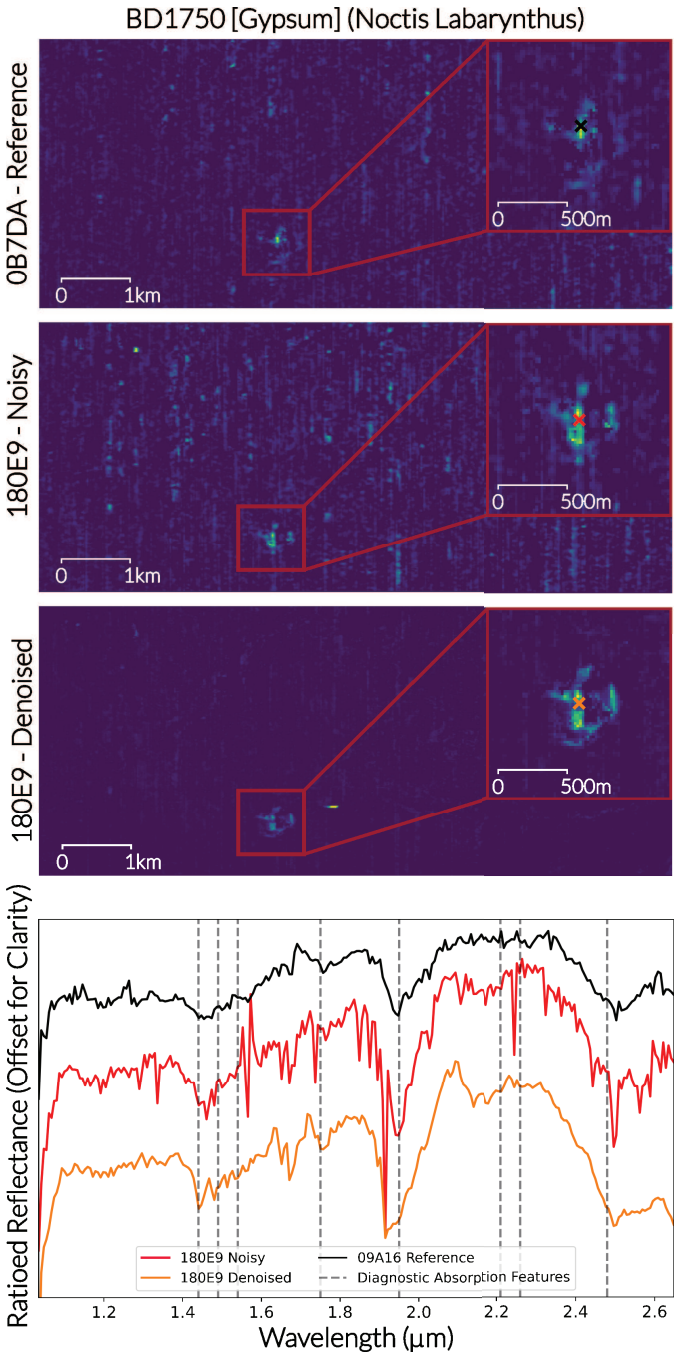


Fig. 6. Case study of two images from Noctis Labarynthus, 0B7DA and 180E9, acquired in 2008 and 2010, respectively. From top to bottom: low-noise reference image, high-noise image, denoised image, and representative spectra. Images highlight absorption features associated with Gypsum, identified in 0B7DA [61]. The red box outlines the area of interest, highlighted in the smaller panel. Spectra shown in the lower panel taken from pixels indicated with crosses, overlain with the absorption features for Gypsum.

The more recently acquired, noisier image in each pair was inputted to N2N4M to denoise and compare it with earlier CRISM images. No pixels from either image were involved in training the model. Standard postprocessing steps were applied to all images, including the ratioing method detailed in Section III-E to highlight uncommon absorption features. To aid in visualization for HSI cubes, summary

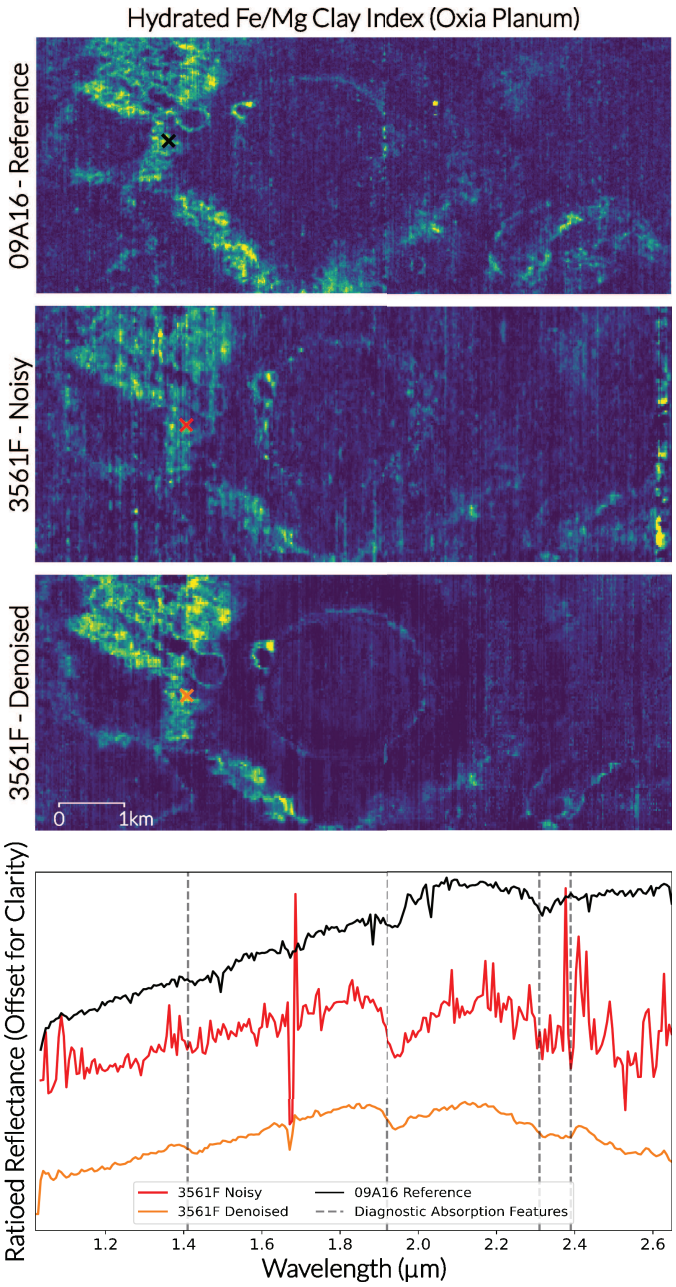


Fig. 7. Case study of two images from Oxia Planum, 09A16 and 3561F, acquired in 2008 and 2015, respectively. From top to bottom: low-noise reference image, high-noise image, denoised image, and representative spectra. Images highlight absorption features associated with hydrated Fe/Mg clays, identified in 09A16 [10]. Spectra shown in the lower panel were taken from pixels indicated with crosses, overlain with the absorption features for hydrated Fe/Mg clays.

parameters from [39] and [58] were calculated to highlight gypsum and hydrated Fe/Mg clays—these mineralogies have been identified in the low-noise images in prior literature [10], [61]. The denoised images (Figs. 6 and 7) exhibit a strong increase in perceptual quality.

In Fig. 6, many small, spurious highlighted regions are removed, leaving the only area positively identified by [61] as containing gypsum. This is a significant increase in perceptual quality, to the level of a high-quality, low-noise image, demonstrating the effectiveness of the proposed method. When

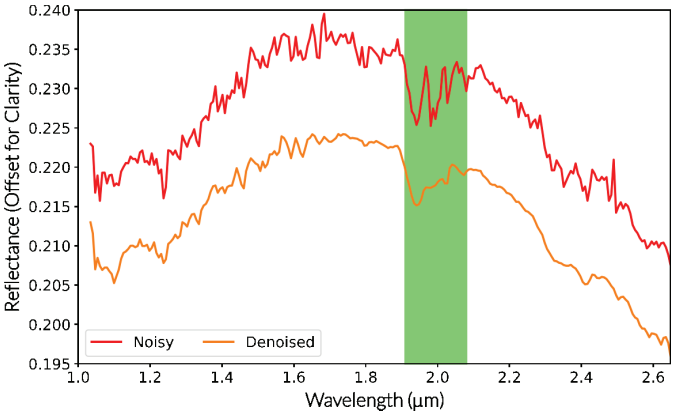


Fig. 8. Noisy spectrum from image FRS0000364CA and denoised prediction from N2N4M. The green zone highlights the region of remnant atmospheric artifact in the noisy spectra, which is removed during denoising.

comparing the selected spectra in the bottom panel of Fig. 6, the denoised spectra show a marked improvement over the noisy spectra. The absorption features align closely with those of the reference spectra, allowing for easy identification of gypsum mineralogy.

In addition to removing additive noise related to the increased sensor temperature, N2N4M shows the ability to mitigate atmospheric artifacts. Such artifacts (see Fig. 8) are remnants, occasionally left after the Volcano-Scan correction, as the atmospheric transmission spectra are not always a perfect fit [8]. It is clear in Fig. 8 that the remnant spikes between 1.9 and 2.1 μm are correctly removed by the model. An additional example can be observed in Fig. 6. These suggest that the model has learned to predict the true shape of the underlying spectra rather than simulating a low-pass filter, which would not remove such effects. Although we do not expect this to allow for identification of precise mineralogy based on that spectral region, it may allow for confirmation of broader features associated with hydrated mineral species.

In the other case study, the denoising process (Fig. 7) has mitigated the striping and edge artifacts evident in the noisy acquisition. Diffuse areas of clay absorptions in the lower section of the image are sharper postdenoising, with particularly strong absorption areas closely resembling those of the reference image. Two crater-rim deposits are also highlighted in the denoised reconstruction, as in the reference, which are obscured in the noisy image. The denoised spectra shown in the bottom panel of Fig. 7 closely resemble that of the reference spectra, with a significantly reduced noise fraction compared to the model input. Critically, no spurious absorption features are introduced during the denoising process, which could prevent correct mineral identification.

V. Conclusion

In this work, a new hyperspectral denoising method was presented, tailored for use on CRISM data. A novel DL architecture, based on the Noise2Noise framework, was adapted for use with hyperspectral imagery. The standard 2-D convolutions were replaced with 1-D convolutions to leverage the large number of channels in CRISM data. This marks CRISM as

unique among other hyperspectral sensors often used for DL studies, which typically have far fewer channels or methods that involve dimensionality reduction before use. This architecture has shown the capability to be trained effectively even with partially noisy target data, which is essential for direct training with CRISM data, where no true "noise-free" data exist. Here, synthetic Gaussian noise was added to the spectra to allow for training without resorting to ground-truth data from other sensors. Further work should experiment with more complex noise distributions that more closely match the noise observed in CRISM data, potentially incorporating the spatial striping artifacts.

We achieve significant performance increases over benchmarks on denoising unseen hyperspectral data. This underscores the importance of training DL methods on the exact sensor for intended use and that transferring may not be effective due to large differences in sensor characteristics. Our model has a significantly smaller parameter count than comparable methods, due to the use of 1-D convolutions (1M as opposed to 1.7M for QRNN3D and 2.7M for Memnet). This is not entirely unexpected, as it has been noted in the literature that smaller, tailored architectures may extract more performance on remote sensing datasets [19].

For the evaluation of the downstream tasks, both N2N4M and CoTCAT outperform the ML benchmarks. CoTCAT is an analytical method based on moving average filters [3], rather than a DL model. This highlights that assessing the impact of new upstream methods on intended downstream tasks is critical for rigorous assessment of these methods. We suggest that the reliance on spatial correlations in noise profiles for the DL benchmark methods may be part of the cause of their poor performance. Pixels in the local area to the spectrum of interest may belong to a different mineralogy, or have significantly different noise levels due to striping artifacts [5]. This may cause these models to remove absorption features that are key for mineral identification as noise, decreasing their downstream task performance. Our future work will investigate the use of spatial correlations in CRISM data to aid the denoising process.

Finally, our methodology is applied to two case study sites, each previously studied for hydrated mineralogy. The results show that our model can increase the visual clarity of summary parameters used to highlight areas of interest. This reduces spurious highlights that require manual review, allowing for easier analysis of CRISM data. Additionally, no false-positive features are introduced, increasing trust in the system, which is essential for use in planetary science and remote sensing.

Appendix
Hyperparameter Tuning

The following hyperparameter space was searched: upsampling/downsampling blocks $\in \{1, 2, 3, 4, 5\}$, batchsize $\in \{32, 64, 128, 256, 512\}$, convolutional kernel size $\in \{3, 5, 7, 9, 11\}$, number of convolutional layers per block $\in \{1, 2, 3\}$, and the learning rate was log uniform sampled between 1×10^{-4} and 1×10^{-1}. Tuning was conducted for 8 h using an Nvidia A100 GPU, with 40 parameter runs completed.

The final model used the following architecture and training regime: four downsampling/upsampling blocks of three convolutional layers each, with a kernel size of 5. A batch size of 256 was used with a learning rate of 0.001 for training of 100 epochs, sufficient to trigger early stopping conditions.

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References

[1] S. L. Murchie et al., "Compact reconnaissance imaging spectrometer for Mars (CRISM) on Mars reconnaissance orbiter (MRO)," *J. Geophys. Res., Planets*, vol. 112, no. E5, 2007, Art. no. E05S03. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1029/2006JE002682

[2] F. P. Seelos et al., "The CRISM investigation in Mars orbit: Overview, history, and delivered data products," *Icarus*, vol. 419, Sep. 2024, Art. no. 115612. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0019103523001896

[3] B. Bultel, C. Quantin, and L. Lozac'h, "Description of CoTCAT (complement to CRISM analysis toolkit)," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3039–3049, Jun. 2015.

[4] J. Brossier et al., "Constraining the spectral behavior of the clay-bearing outcrops in oxia planum, the landing site for ExoMars 'Rosalind Franklin' rover," *Icarus*, vol. 386, Nov. 2022, Art. no. 115114. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0019103522002214

[5] Y. Itoh and M. Parente, "A new method for atmospheric correction and de-noising of CRISM hyperspectral data," *Icarus*, vol. 354, Jan. 2021, Art. no. 114024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0019103520303857

[6] J. Carter, F. Poulet, J. Bibring, N. Mangold, and S. L. Murchie, "Hydrous minerals on Mars as seen by the CRISM and OMEGA imaging spectrometers: Updated global view," *J. Geophys. Res., Planets*, vol. 118, no. 4, pp. 831–858, Apr. 2012. [Online]. Available: https://onlinelibrary.wiley.com/doi/pdf/10.1029/2012JE004145

[7] P. E. Martin and E. F. Barker, "The infrared absorption spectrum of carbon dioxide," *Phys. Rev.*, vol. 41, no. 3, pp. 291–303, Aug. 1932. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRev.41.291

[8] P. C. McGuire et al., "An improvement to the volcano-scan algorithm for atmospheric correction of CRISM and OMEGA spectral data," *Planet. Space Sci.*, vol. 57, no. 7, pp. 809–815, Jun. 2009. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0032063309000798

[9] J. Lever et al., "Facing & mitigating common challenges when working with real-world data: The data learning paradigm," *J. Comput. Sci.*, vol. 85, Feb. 2025, Art. no. 102523. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877750324003168

[10] L. Mandon et al., "Morphological and spectral diversity of the clay-bearing unit at the ExoMars landing site oxia planum," *Astrobiology*, vol. 21, no. 4, pp. 464–480, Apr. 2021. [Online]. Available: https://www.liebertpub.com/doi/10.1089/ast.2020.2292

[11] M. Fernandes, A. Pletl, N. Thomas, A. P. Rossi, and B. Elser, "Generation and optimization of spectral cluster maps to enable data fusion of CaSSIS and CRISM datasets," *Remote Sens.*, vol. 14, no. 11, p. 2524, May 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/11/2524

[12] A. Pletl, M. Fernandes, N. Thomas, A. P. Rossi, and B. Elser, "Spectral clustering of CRISM datasets in Jezero Crater using UMAP and k-means," *Remote Sens.*, vol. 15, no. 4, p. 939, Feb. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/4/939

[13] N. Thomas et al., "The colour and stereo surface imaging system (CaSSIS) for the ExoMars trace gas orbiter," *Space Sci. Rev.*, vol. 212, nos. 3–4, pp. 1897–1944, Nov. 2017, doi: 10.1007/s11214-017-0421-1.

[14] A. Koeppel, H. Eifert, and A. D. Rogers, "Building a Mars surface classification model from analog thermal, spectral, and physical data," in *Proc. 10th Int. Conf. Mars*, 2024.

[15] A. M. Saranathan and M. Parente, "Adversarial feature learning for improved mineral mapping of CRISM data," *Icarus*, vol. 355, Feb. 2021, Art. no. 114107. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0019103520304528

[16] Y. Tao, S. J. Conway, J.-P. Müller, A. R. D. Putri, N. Thomas, and G. Cremonese, "Single image super-resolution restoration of TGO CaSSIS colour images: Demonstration with perseverance rover landing site and Mars science targets," *Remote Sens.*, vol. 13, no. 9, p. 1777, May 2021. [Online]. Available: https://www.mdpi.com/2072-4292/13/9/1777

[17] E. Plebani, B. L. Ehlmann, E. K. Leask, V. K. Fox, and M. M. Dundar, "A machine learning toolkit for CRISM image analysis," *Icarus*, vol. 376, Apr. 2022, Art. no. 114849. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0019103521004905

[18] P. Kumari, S. Soor, A. Shetty, and S. G. Koolagudi, "MICAnet: A deep convolutional neural network for mineral identification on Martian surface," *Egyptian J. Remote Sens. Space Sci.*, vol. 27, no. 3, pp. 501–507, Sep. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1110982324000474

[19] E. Rolf, K. Klemmer, C. Robinson, and H. Kerner, "Position: Mission critical satellite data is a distinct modality in machine learning," in *Proc. 41st Int. Conf. Mach. Learn.*, Jul. 2024, pp. 42691–42706. [Online]. Available: https://proceedings.mlr.press/v235/rolf24a.html

[20] S. L. Murchie et al., "Compact reconnaissance imaging spectrometer for Mars investigation and data set from the Mars reconnaissance Orbiter's primary science phase," *J. Geophys. Res., Planets*, vol. 114, no. E2, Feb. 2009, Art. no. E00D07. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1029/2009JE003344

[21] F. P. Seelos, S. L. Murchie, D. C. Humm, O. S. Barnouin, and F. Mor, "Data processing and analysis products update—Calibration, correction, and visualisation," in *Proc. 42nd Lunar Planet. Sci. Conf.*, 2011.

[22] S. Murchie, E. A. Guinness, and S. Slavney, "CRISM data product software interface specification version 1.3.7.6," NASA/Jet Propulsion Laboratory, Pasadena, CA, USA, Tech. Rep. JPL D 12345, Version 1.3.7.7, Jun. 2022.

[23] J. F. Mustard et al., "Hydrated silicate minerals on Mars observed by the Mars reconnaissance orbiter CRISM instrument," *Nature*, vol. 454, no. 7202, pp. 305–309, Jul. 2008. [Online]. Available: https://www.nature.com/articles/nature07097

[24] C. Quantin-Nataf et al., "MarsSI: Martian surface data processing information system," *Planet. Space Sci.*, vol. 150, pp. 157–170, Jan. 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0032063316304718

[25] M. Parente, "A new approach to denoise CRISM images," in *Proc. 39th Lunar Planet. Sci. Conf.*, Mar. 2008. [Online]. Available: https://www.lpi.usra.edu/meetings/lpsc2008/pdf/2528.pdf

[26] J. Carter, F. Poulet, S. Murchie, and J. P. Bibring, "Automated processing of planetary hyperspectral datasets for the extraction of weak mineral signatures and applications to CRISM observations of hydrated silicates on Mars," *Planet. Space Sci.*, vol. 76, pp. 53–67, Feb. 2013. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0032063312003625

[27] P. C. McGuire et al., "MRO/CRISM retrieval of surface Lambert albedos for multispectral mapping of Mars with DISORT-based radiative transfer modeling: Phase 1—Using historical climatology for temperatures, aerosol optical depths, and atmospheric pressures," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 12, pp. 4020–4040, Dec. 2008. [Online]. Available: http://ieeexplore.ieee.org/document/4685531/

[28] C. D. Kreisch et al., "Regularization of Mars reconnaissance orbiter CRISM along-track oversampled hyperspectral imaging observations of Mars," *Icarus*, vol. 282, pp. 136–151, Jan. 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0019103516301579

[29] L. He, J. A. O'Sullivan, D. V. Politte, K. E. Powell, and R. E. Arvidson, "Quantitative reconstruction and denoising method HyBER for hyperspectral image data and its application to CRISM," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1219–1230, Apr. 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8672185

[30] Q. Zhang, Y. Zheng, Q. Yuan, M. Song, H. Yu, and Y. Xiao, "Hyperspectral image denoising: From model-driven, data-driven, to model-data-driven," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 1–21, Jun. 2023, Art. no. 3278866. [Online]. Available: https://ieeexplore.ieee.org/document/10144690/arnumber=10144690

[31] B. Lin, X. Tao, X. Qin, Y. Duan, and J. Lu, "Hyperspectral image denoising via nonnegative matrix factorization and convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 4023–4026. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8517388

[32] Y. Chang, L. Yan, H. Fang, S. Zhong, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8435923

[33] W. Liu and J. Lee, "A 3-D atrous convolution neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5701–5715, Aug. 2019. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8676115#citations

[34] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8454887

[35] Y. Tai, J. Yang, X. Liu, and C. Xu, "MemNet: A persistent memory network for image restoration," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 4549–4557. [Online]. Available: http://ieeexplore.ieee.org/document/8237748/

[36] K. Wei, Y. Fu, and H. Huang, "3D quasi-recurrent neural network for hyperspectral image denoising," 2020, *arXiv:2003.04547*.

[37] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," 2015, *arXiv:1505.04597*.

[38] J. Lehtinen et al., "Noise2Noise: Learning image restoration without clean data," 2018, *arXiv:1803.04189*.

[39] C. E. Viviano et al., "Revised CRISM spectral parameters and summary products based on the currently detected mineral diversity on Mars," *J. Geophys. Res., Planets*, vol. 119, no. 6, pp. 1403–1431, Jun. 2014. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/2014JE004627

[40] P. Kumari, S. Soor, A. Shetty, and S. G. Koolagudi, "A fully-automated framework for mineral identification on Martian surface using supervised learning models," *IEEE Access*, vol. 11, pp. 13121–13137, 2023. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10038675

[41] F. He, T. Liu, and D. Tao, "Control batch size and learning rate to generalize well: Theoretical and empirical evidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1141–1150.

[42] L. Sun, J. Zhang, J. Li, Y. Wang, and D. Zeng, "SDFC dataset: A large-scale benchmark dataset for hyperspectral image classification," *Opt. Quantum Electron.*, vol. 55, no. 2, p. 173, Jan. 2023, doi: 10.1007/s11082-022-04399-9.

[43] S. Bickel, M. Brückner, and T. Scheffer, "Discriminative learning under covariate shift," *J. Mach. Learn.*, vol. 10, no. 75, pp. 2137–2155, 2009.

[44] H. Meyer and E. Pebesma, "Machine learning-based global maps of ecological variables and the challenge of assessing them," *Nature Commun.*, vol. 13, no. 1, p. 2208, Apr. 2022. [Online]. Available: https://www.nature.com/articles/s41467-022-29838-9

[45] K. Pawluszek-Filipiak and A. Borkowski, "On the importance of train–test split ratio of datasets in automatic landslide detection by supervised classification," *Remote Sens.*, vol. 12, no. 18, p. 3054, Sep. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/18/3054

[46] R. Thoreau, L. Risser, V. Achard, B. Berthelot, and X. Briottet, "Toulouse hyperspectral data set: A benchmark data set to assess semi-supervised spectral representation learning and pixel-wise classification techniques," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 323–337, Jun. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271624002004

[47] P. Gomez, E. V. Bohacek, and R. S. Bahia, "Martian digital elevation models derived from monocular depth estimation," in *Proc. 10th Int. Conf. Mars*, 2024.

[48] B. Kalantar, B. Pradhan, S. A. Naghibi, A. Motevalli, and S. Mansor, "Assessment of the effects of training data selection on the landslide susceptibility mapping: A comparison between support vector machine (SVM), logistic regression (LR) and artificial neural networks (ANN)," *Geomatics, Natural Hazards Risk*, vol. 9, no. 1, pp. 49–69, Jan. 2018, doi: 10.1080/19475705.2017.1407368.

[49] K. Ayush et al., "Geography-aware self-supervised learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10161–10170. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/AyushGeography-AwareSelf-SupervisedLearningICCV2021paper.html

[50] W. Chen, X. Li, Y. Wang, G. Chen, and S. Liu, "Forested landslide detection using LiDAR data and the random forest algorithm: A case study of the three Gorges, China," *Remote Sens. Environ.*, vol. 152, pp. 291–301, Sep. 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425714002491

[51] S. Georganos et al., "Geographical random forests: A spatial extension of the random forest algorithm to address spatial heterogeneity in remote sensing and population modelling," *Geocarto Int.*, vol. 36, no. 2, pp. 121–136, Jan. 2021, doi: 10.1080/10106049.2019.1595177.

[52] J. J. Salazar, L. Garland, J. Ochoa, and M. J. Pyrcz, "Fair train-test split in machine learning: Mitigating spatial autocorrelation for improved prediction accuracy," *J. Petroleum Sci. Eng.*, vol. 209, Feb. 2022, Art. no. 109885. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0920410521015023

[53] M. Parente, H. D. Makarewicz, and J. L. Bishop, "Decomposition of mineral absorption bands using nonlinear least squares curve fitting: Application to Martian meteorites and CRISM data," *Planet. Space Sci.*, vol. 59, nos. 5–6, pp. 423–442, Apr. 2011. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0032063311000377

[54] M. Parente, J. T. Clark, A. J. Brown, and J. L. Bishop, "End-to-end simulation and analytical model of remote-sensing systems: Application to CRISM," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 3877–3888, Nov. 2010.

[55] D. Coquelin, B. Rasti, M. Gotz, P. Ghamisi, R. Gloaguen, and A. Streit, "Hyde: The first open-source, Python-based, GPU-accelerated hyperspectral denoising package," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–5.

[56] P. A. Gorry, "General least-squares smoothing and differentiation by the convolution (Savitzky–Golay) method," *Anal. Chem.*, vol. 62, no. 6, pp. 570–573, Mar. 1990, doi: 10.1021/ac00205a007.

[57] B. L. Ehlmann et al., "Identification of hydrated silicate minerals on Mars using MRO-CRISM: Geologic context near Nili fossae and implications for aqueous alteration," *J. Geophys. Res., Planets*, vol. 114, no. E2, Feb. 2009, Art. no. E00D08. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1029/2009JE003339

[58] S. M. Pelkey et al., "CRISM multispectral summary products: Parameterizing mineral diversity on Mars from reflectance," *J. Geophys. Res., Planets*, vol. 112, no. E8, 2007, Art. no. E08S14. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1029/2006JE002831

[59] B. L. Ehlmann, J. F. Mustard, and S. L. Murchie, "Geologic setting of serpentine deposits on Mars," *Geophys. Res. Lett.*, vol. 37, no. 6, Mar. 2010, Art. no. L06201. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1029/2010GL042596

[60] A. J. Brown, "Spectral curve fitting for automatic hyperspectral data analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1601–1608, Jun. 2006. [Online]. Available: https://ieeexplore.ieee.org/document/1634723

[61] P. Thollot, "Etude de l'altration de la surface de Mars par imagerie hyperspectrale: Minralogie design rgions de noctis labyrinthus et valles marineris," Ph.D. dissertation, Doctoral School Eng. Sci., Geosci., Archit., Univ. Nantes, Nantes, France, Jan. 2013. [Online]. Available: https://theses.fr/2013NANT2076



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